AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	9.1 / 1.63243 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-03 10:00 UTC
Location	29.5831, -95.0155
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	6 / 421	50.67 to 100; mean 83.4543	91.32 at 2026-06-03 09:55Z; 5 min before event
asos	Air temperature (temperature)	degC	6 / 421	22 to 33; mean 26.2482	23.5 at 2026-06-03 09:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	6 / 420	0 to 360; mean 90.7619	20 at 2026-06-03 09:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	6 / 421	0 to 9.77444; mean 2.9083	5.65888 at 2026-06-03 09:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	6 / 72	5866.59 to 5898.5; mean 5879.83	5868.02 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	6 / 72	14.9006 to 1735.86; mean 447.918	231.159 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	6 / 72	46.2607 to 56.7462; mean 51.46	53.9548 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	6 / 72	1009.78 to 1015.86; mean 1013.02	1014.81 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	6 / 72	15.9888 to 19.4844; mean 17.4545	16.7966 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	6 / 72	-4.66046 to 0.147017; mean -2.2013	-2.41627 at 2026-06-03 12:00Z; 2 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	6 / 72	-3.20242 to 3.40106; mean 0.1268	-1.09242 at 2026-06-03 12:00Z; 2 h after event
openaq	Ozone (ozone)	ppm	11 / 747	-0.002 to 0.094; mean 0.0182	0.016 at 2026-06-03 10:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	2 / 145	9 to 78; mean 20.8207	24 at 2026-06-03 10:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	24 / 2744	-3.9 to 42.5; mean 5.9601	8.1 at 2026-06-03 10:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	3 / 216	0 to 100; mean 66.6991	76 at 2026-06-03 09:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	3 / 216	63 to 97; mean 84.2454	93 at 2026-06-03 09:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	3 / 216	0 to 9.99; mean 0.3263	0 at 2026-06-03 09:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	3 / 216	1012 to 1017; mean 1014.67	1015 at 2026-06-03 09:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	3 / 216	22.69 to 33.69; mean 26.4841	24.66 at 2026-06-03 09:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	3 / 216	0 to 345; mean 103.204	32 at 2026-06-03 09:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	3 / 216	0 to 6.28; mean 2.4649	2.12 at 2026-06-03 09:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	53 / 3869	0 to 40.4; mean 6.6842	5.2 at 2026-06-03 10:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	5 / 5	0.0277457 to 0.0294826; mean 0.0285	0.0294826 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	3 / 3	0.000135539 to 0.000169173; mean 0.0001	0.000169173 at 2026-06-02 19:52Z; 14 h 8 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	4.04755e-05 to 4.7738e-05; mean 0	4.04755e-05 at 2026-06-02 19:52Z; 14 h 8 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	4 / 4	-9.61344e-06 to 1.55388e-06; mean -0	-9.16616e-06 at 2026-06-02 19:52Z; 14 h 8 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-03 19:04Z; 9 h 4 min after event
tceq	Carbon monoxide (co)	ppm	3 / 217	0 to 0.7; mean 0.1783	0.2 at 2026-06-03 10:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	11 / 770	-2.6 to 28.2; mean 7.4483	9.1 at 2026-06-03 10:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 214	-0.6 to 3.6; mean -0.0084	0.5 at 2026-06-03 10:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-01 18Z 66.97, 06-02 00Z 82.71, 06Z 92.61, 12Z 75.68, 18Z 63.29, 06-03 00Z 82.43, 06Z 87.99, 12Z 80.59, 18Z 89.39, 06-04 00Z 92.78, 06Z 95.19, 12Z 87.71, 18Z 75.79
asos	Air temperature (temperature)	06-01 18Z 30.48, 06-02 00Z 27.5, 06Z 25.48, 12Z 28.72, 18Z 30.68, 06-03 00Z 25.09, 06Z 24.18, 12Z 26.26, 18Z 24.79, 06-04 00Z 23.94, 06Z 23.64, 12Z 25.54, 18Z 27.99
asos	Wind direction (wind_direction)	06-01 18Z 154.2, 06-02 00Z 135.9, 06Z 39.7, 12Z 89.44, 18Z 108.1, 06-03 00Z 80, 06Z 154.1, 12Z 81.71, 18Z 71.71, 06-04 00Z 69.72, 06Z 70.59, 12Z 60.28, 18Z 114.2
asos	Wind speed (wind_speed)	06-01 18Z 4.673, 06-02 00Z 2.209, 06Z 0.4365, 12Z 2.086, 18Z 4.216, 06-03 00Z 4.468, 06Z 3.041, 12Z 3.487, 18Z 2.705, 06-04 00Z 2.715, 06Z 1.952, 12Z 2.244, 18Z 5.294
noaa_gfs	500-hPa geopotential height (gh_500)	06-02 00Z 5889, 06Z 5898, 12Z 5881, 18Z 5893, 06-03 00Z 5879, 06Z 5880, 12Z 5868, 18Z 5879, 06-04 00Z 5868, 06Z 5875, 12Z 5869, 18Z 5879
noaa_gfs	Planetary boundary layer height (pbl_height)	06-02 00Z 586.2, 06Z 90.71, 12Z 51.14, 18Z 1386, 06-03 00Z 636.9, 06Z 249.6, 12Z 186, 18Z 711.2, 06-04 00Z 274.2, 06Z 152.8, 12Z 129.1, 18Z 921.1
noaa_gfs	Precipitable water (precipitable_water)	06-02 00Z 50.87, 06Z 50.8, 12Z 51.26, 18Z 50.72, 06-03 00Z 48.19, 06Z 50.34, 12Z 52.98, 18Z 53.94, 06-04 00Z 51.95, 06Z 51.69, 12Z 50.62, 18Z 54.16
noaa_gfs	Surface pressure (surface_pressure)	06-02 00Z 1011, 06Z 1013, 12Z 1013, 18Z 1013, 06-03 00Z 1011, 06Z 1013, 12Z 1014, 18Z 1015, 06-04 00Z 1013, 06Z 1014, 12Z 1013, 18Z 1013
noaa_gfs	850-hPa temperature (t_850)	06-02 00Z 19.2, 06Z 18.52, 12Z 17.52, 18Z 16.92, 06-03 00Z 19.13, 06Z 18.73, 12Z 16.72, 18Z 16.33, 06-04 00Z 16.7, 06Z 16.72, 12Z 16.49, 18Z 16.47
noaa_gfs	10-m eastward wind (u_10m)	06-02 00Z -2.557, 06Z -0.6094, 12Z -0.713, 18Z -2.171, 06-03 00Z -2.616, 06Z -2.736, 12Z -1.501, 18Z -3.684, 06-04 00Z -2.205, 06Z -2.292, 12Z -1.886, 18Z -3.446
noaa_gfs	10-m northward wind (v_10m)	06-02 00Z 2.939, 06Z 1.462, 12Z -0.8894, 18Z -1.045, 06-03 00Z 2.048, 06Z 0.7416, 12Z -2.342, 18Z -0.9045, 06-04 00Z 0.06962, 06Z -0.309, 12Z -0.1891, 18Z -0.05956
openaq	Ozone (ozone)	06-01 18Z 0.0409, 06-02 00Z 0.01737, 06Z 0.007267, 12Z 0.0102, 18Z 0.0319, 06-03 00Z 0.02336, 06Z 0.0208, 12Z 0.02065, 18Z 0.0207, 06-04 00Z 0.01297, 06Z 0.01095, 12Z 0.01184, 18Z 0.02391
openaq	PM10 (pm10)	06-01 18Z 32, 06-02 00Z 25.17, 06Z 22.08, 12Z 31.17, 18Z 28.42, 06-03 00Z 17.5, 06Z 15.83, 12Z 22.08, 18Z 15.27, 06-04 00Z 18.33, 06Z 20.67, 12Z 15.08, 18Z 12.7
openaq	PM2.5 (pm25)	06-01 18Z 6.282, 06-02 00Z 5.418, 06Z 5.072, 12Z 6.814, 18Z 5.072, 06-03 00Z 5.073, 06Z 4.677, 12Z 7.239, 18Z 5.366, 06-04 00Z 6.7, 06Z 8.49, 12Z 7.995, 18Z 4.127
openweather	Cloud cover (cloud_cover)	06-01 18Z 33.33, 06-02 00Z 14.44, 06Z 27.5, 12Z 60.89, 18Z 85.61, 06-03 00Z 30.06, 06Z 87.39, 12Z 99.5, 18Z 86.89, 06-04 00Z 98.11, 06Z 98.33, 12Z 69.39, 18Z 46.75
openweather	Relative humidity (humidity)	06-01 18Z 72.33, 06-02 00Z 80.28, 06Z 91.17, 12Z 79.17, 18Z 70.67, 06-03 00Z 83.72, 06Z 85.83, 12Z 84.39, 18Z 85.11, 06-04 00Z 91.67, 06Z 93.17, 12Z 88.72, 18Z 79.42
openweather	Precipitation rate (precipitation)	06-01 18Z 0, 06-02 00Z 0, 06Z 0, 12Z 0, 18Z 0.65, 06-03 00Z 0.1339, 06Z 0, 12Z 0.8372, 18Z 1.733, 06-04 00Z 0, 06Z 0.08389, 12Z 0.4778, 18Z 0
openweather	Surface pressure (pressure)	06-01 18Z 1012, 06-02 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1013, 06-03 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 06-04 00Z 1015, 06Z 1015, 12Z 1015, 18Z 1014
openweather	Air temperature (temperature)	06-01 18Z 30.88, 06-02 00Z 28.14, 06Z 25.79, 12Z 29.07, 18Z 31.35, 06-03 00Z 24.93, 06Z 24.07, 12Z 26.5, 18Z 25.84, 06-04 00Z 23.87, 06Z 23.41, 12Z 25.5, 18Z 28.58
openweather	Wind direction (wind_direction)	06-01 18Z 150, 06-02 00Z 166.7, 06Z 52.22, 12Z 126, 18Z 138.2, 06-03 00Z 80.06, 06Z 56.94, 12Z 48.61, 18Z 144.4, 06-04 00Z 82.83, 06Z 124, 12Z 86.28, 18Z 123.2
openweather	Wind speed (wind_speed)	06-01 18Z 4.973, 06-02 00Z 3.194, 06Z 0.9728, 12Z 2.13, 18Z 2.486, 06-03 00Z 2.476, 06Z 1.785, 12Z 2.639, 18Z 3.984, 06-04 00Z 2.277, 06Z 1.444, 12Z 2.501, 18Z 3.046

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-01 18Z 5.691, 06-02 00Z 5.247, 06Z 5.65, 12Z 6.641, 18Z 4.326, 06-03 00Z 5.928, 06Z 6.473, 12Z 8.26, 18Z 5.821, 06-04 00Z 8.315, 06Z 11.37, 12Z 8.648, 18Z 3.293
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	CO column (s5p_co_column)	06-02 18Z 0.02882, 06-03 18Z 0.02948, 06-04 18Z 0.02776
sentinel5p	CO granules available (s5p_co_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-02 18Z 0.0001692, 06-04 18Z 0.0001378
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-02 18Z 4.07e-05, 06-04 18Z 4.712e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-02 18Z -8.155e-06, 06-04 18Z -4.03e-06
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-02 18Z 1, 06-03 18Z 1, 06-04 18Z 1
tceq	Carbon monoxide (co)	06-01 18Z 0.2667, 06-02 00Z 0.2, 06Z 0.2333, 12Z 0.2056, 18Z 0.1667, 06-03 00Z 0.1389, 06Z 0.1222, 12Z 0.1667, 18Z 0.2, 06-04 00Z 0.1611, 06Z 0.1444, 12Z 0.1882, 18Z 0.1867
tceq	Nitrogen dioxide (no2)	06-01 18Z 4.859, 06-02 00Z 4.542, 06Z 5.12, 12Z 7.794, 18Z 6.039, 06-03 00Z 5.923, 06Z 6.363, 12Z 8.881, 18Z 12.74, 06-04 00Z 9.767, 06Z 8.739, 12Z 9.252, 18Z 4.402
tceq	Sulfur dioxide (so2)	06-01 18Z 0.03333, 06-02 00Z -0.1444, 06Z -0.1294, 12Z -0.01176, 18Z 0.1556, 06-03 00Z -0.01667, 06Z -0.02778, 12Z 0.55, 18Z 0.1313, 06-04 00Z -0.2056, 06Z -0.2556, 12Z -0.1529, 18Z -0.006667

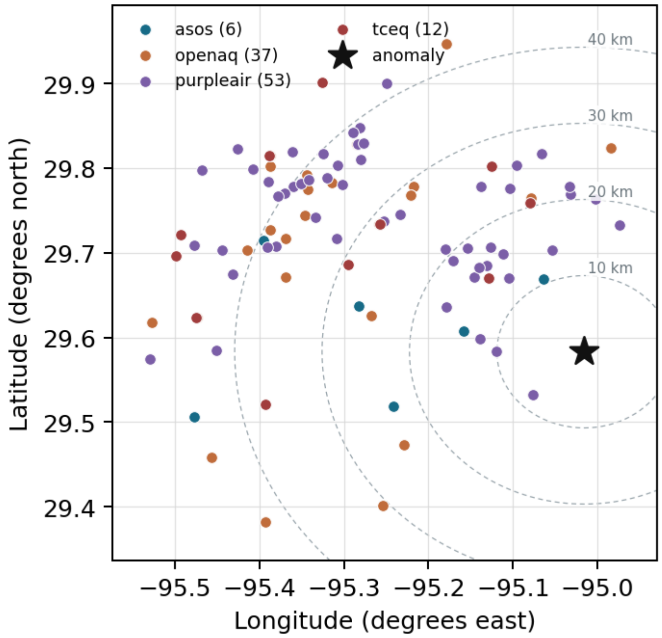
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

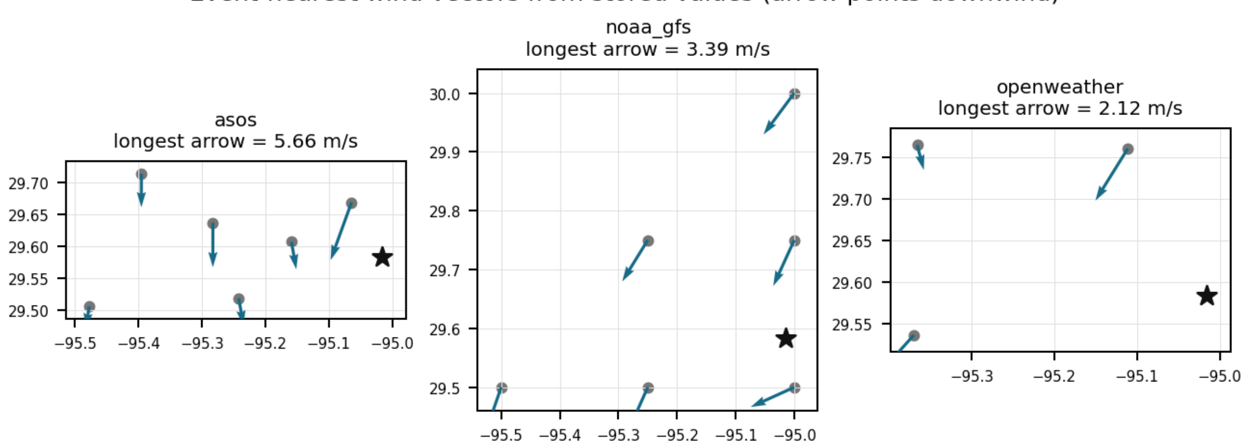
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (10.667 km) 85.48, EFD (14.102 km) 86.39, LVJ (23.01 km) 84.18, HOU (26.499 km) 82.06, MCJ (39.455 km) 84.61, AXH (45.443 km) 78.38
asos	Air temperature (temperature)	T41 (10.667 km) 26.14, EFD (14.102 km) 26.23, LVJ (23.01 km) 26.21, HOU (26.499 km) 26.43, MCJ (39.455 km) 25.93, AXH (45.443 km) 26.53
asos	Wind direction (wind_direction)	T41 (10.667 km) 93.61, EFD (14.102 km) 87.34, LVJ (23.01 km) 88.19, HOU (26.499 km) 98.47, MCJ (39.455 km) 105.7, AXH (45.443 km) 71.67
asos	Wind speed (wind_speed)	T41 (10.667 km) 3.08, EFD (14.102 km) 2.934, LVJ (23.01 km) 2.351, HOU (26.499 km) 3.337, MCJ (39.455 km) 3.944, AXH (45.443 km) 1.851
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.00 (9.357 km) 5880, gfs:29.75,-95.00 (18.624 km) 5879, gfs:29.50,-95.25 (24.49 km) 5880, gfs:29.75,-95.25 (29.288 km) 5880, gfs:30.00,-95.00 (46.387 km) 5879, gfs:29.50,-95.50 (47.768 km) 5881
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.00 (9.357 km) 466.1, gfs:29.75,-95.00 (18.624 km) 388.6, gfs:29.50,-95.25 (24.49 km) 429.1, gfs:29.75,-95.25 (29.288 km) 524.1, gfs:30.00,-95.00 (46.387 km) 433, gfs:29.50,-95.50 (47.768 km) 446.5
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.00 (9.357 km) 52.09, gfs:29.75,-95.00 (18.624 km) 52.13, gfs:29.50,-95.25 (24.49 km) 51.49, gfs:29.75,-95.25 (29.288 km) 51.09, gfs:30.00,-95.00 (46.387 km) 51.09, gfs:29.50,-95.50 (47.768 km) 50.87

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.00 (9.357 km) 1014, gfs:29.75,-95.00 (18.624 km) 1014, gfs:29.50,-95.25 (24.49 km) 1013, gfs:29.75,-95.25 (29.288 km) 1013, gfs:30.00,-95.00 (46.387 km) 1012, gfs:29.50,-95.50 (47.768 km) 1012
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.00 (9.357 km) 17.42, gfs:29.75,-95.00 (18.624 km) 17.43, gfs:29.50,-95.25 (24.49 km) 17.5, gfs:29.75,-95.25 (29.288 km) 17.54, gfs:30.00,-95.00 (46.387 km) 17.33, gfs:29.50,-95.50 (47.768 km) 17.49
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.00 (9.357 km) -2.787, gfs:29.75,-95.00 (18.624 km) -2.066, gfs:29.50,-95.25 (24.49 km) -2.113, gfs:29.75,-95.25 (29.288 km) -1.995, gfs:30.00,-95.00 (46.387 km) -2.077, gfs:29.50,-95.50 (47.768 km) -2.17
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.00 (9.357 km) 0.7847, gfs:29.75,-95.00 (18.624 km) 0.388, gfs:29.50,-95.25 (24.49 km) -0.06115, gfs:29.75,-95.25 (29.288 km) -0.08365, gfs:30.00,-95.00 (46.387 km) -0.03115, gfs:29.50,-95.50 (47.768 km) -0.2362
openaq	Ozone (ozone)	332 (14.586 km) 0.02133, 2800 (21.051 km) 0.01769, 320 (24.786 km) 0.01579, 339 (26.892 km) 0.02267, 2039 (28.563 km) 0.01499, 3378 (28.774 km) 0.01603, 325 (29.328 km) 0.01577, 2046 (37.626 km) 0.01608, +3 more
openaq	PM10 (pm10)	271 (14.586 km) 18.77, 13380082 (28.774 km) 22.9
openaq	PM2.5 (pm25)	268 (14.586 km) 7.632, 11638043 (23.95 km) 2.095, 2040 (28.563 km) 8.49, 3379 (28.774 km) 6.499, 8967479 (28.83 km) 6.829, 13787195 (29.199 km) 7.126, 16026701 (30.669 km) 1.359, 9416627 (35.586 km) 4.338, +16 more
openweather	Cloud cover (cloud_cover)	grid:east (21.774 km) 66.47, grid:south (34.665 km) 63.71, grid:center (39.395 km) 69.92
openweather	Relative humidity (humidity)	grid:east (21.774 km) 87.78, grid:south (34.665 km) 81.89, grid:center (39.395 km) 83.07
openweather	Precipitation rate (precipitation)	grid:east (21.774 km) 0.43, grid:south (34.665 km) 0.265, grid:center (39.395 km) 0.284
openweather	Surface pressure (pressure)	grid:east (21.774 km) 1015, grid:south (34.665 km) 1015, grid:center (39.395 km) 1015
openweather	Air temperature (temperature)	grid:east (21.774 km) 26.39, grid:south (34.665 km) 26.52, grid:center (39.395 km) 26.54
openweather	Wind direction (wind_direction)	grid:east (21.774 km) 91.15, grid:south (34.665 km) 103.9, grid:center (39.395 km) 114.6
openweather	Wind speed (wind_speed)	grid:east (21.774 km) 2.791, grid:south (34.665 km) 2.266, grid:center (39.395 km) 2.338
purpleair	PM2.5 (pm25)	2386 (8.161 km) 4.258, 3298 (10.11 km) 7.014, 247283 (12.095 km) 7.036, 268165 (12.925 km) 7.238, 222815 (13.892 km) 5.801, 222593 (15.865 km) 6.996, 128007 (15.888 km) 2.716, 161689 (15.943 km) 6.895, +45 more
tceq	Carbon monoxide (co)	482011039 (14.566 km) 0.1288, 482011035 (28.771 km) 0.1681, 482011052 (44.211 km) 0.2389
tceq	Nitrogen dioxide (no2)	482011050 (0.0 km) 2.858, 482011039 (14.566 km) 7.578, 482011015 (20.507 km) 7.138, 482010026 (26.633 km) 8.527, 482011035 (28.771 km) 8.301, 482010416 (29.325 km) 10.6, 480391004 (37.123 km) 5.981, 482011052 (44.211 km) 8.582, +3 more
tceq	Sulfur dioxide (so2)	482011039 (14.566 km) 0.2639, 482011035 (28.771 km) -0.07945, 482010051 (44.59 km) -0.2174

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to an accumulation of pollutants in the area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

Boundary-layer suppression likely helped elevate near-surface NO2 because humidity near the anomaly was very high at about 91 to 93 percent, cloud cover near the anomaly was 76 percent, and GFS planetary boundary layer height was low at about 231 m near 12Z on 2026-06-03, although the evidence is weakened by observed surface wind speeds of about 5.66 m/s at the nearest ASOS station and 2.12 m/s in openweather that indicate ventilation was not extremely stagnant.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

A fresh local combustion plume from nearby traffic, diesel activity, or industrial combustion is a plausible cause of the NO_2 anomaly because the NO_2 spike occurred without corresponding increases in ozone, PM2.5, PM10, CO, or SO_2, which is consistent with a spatially narrow and recently emitted NOx-rich plume near the monitor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

Co-pollutant and regional-context data show only modest CO, SO_2, PM2.5, PM10, and ozone near the anomaly time, which makes a broad regional episode or a large sulfur-rich combustion event less likely than a localized fresh combustion source.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

The tceq NO2 anomaly occurred at 2026-06-03T10:00:00+00:00 at location (29.583, -95.016).

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The PM2.5 levels are elevated at 5.2 ug/m3, which is higher than the mean of 6.6842 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

ASOS surface meteorological observations recorded light wind speeds around 3.0 m/s alongside high relative humidity near 91 percent at 09:55 UTC, limiting horizontal dispersion near the TCEQ monitoring location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The tceq NO2 anomaly had a z-score of 6.618576381977915 and was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

At 10:00 UTC on June 3, 2026, a TCEQ monitoring station located at coordinates (29.583, -95.016) recorded a severe NO_2 concentration anomaly of 9.1 ppb, compared to an expected baseline value of 1.63 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

NOAA GFS model estimates showed a shallow planetary boundary layer height dropping below 250 meters during the early morning hours of June 3, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

NOAA GFS model data indicates planetary boundary layer heights collapsing below 250 meters during the morning of June 3, 2026, restricting vertical atmospheric mixing and trapping surface-level NO_2 emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

Across the regional TCEQ monitoring network, 6-hour average NO2 concentrations rose significantly on June 3, 2026, reaching a peak mean of 12.74 ppb between 12:00 and 18:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The air quality anomaly in Houston, Texas on June 3rd was caused by a combination of factors including high PM2.5 levels and moderate wind speeds.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

Light north-northeasterly surface winds advected combustion emissions from industrial and transportation corridors toward the coastal TCEQ monitoring site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The moderate wind speeds contributed to the spread and persistence of the air quality anomaly, allowing pollutants to linger in the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

OpenWeather data shows cloud cover exceeding 87 percent in the morning of June 3, 2026, suppressing the photolytic degradation mechanisms that typically deplete ambient NO_2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

ASOS weather stations in the Houston region observed light surface wind speeds averaging around 3.0 m/s and high relative humidity near 88% around 10:00 UTC on June 3, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

A local fresh combustion plume near the anomalous TCEQ NO_2 monitor is supported by the monitor-specific contrast because the anomalous monitor measured 9.1 ppb at 2026-06-03T10:00:00+00:00 against a site mean of 2.858 ppb while the network-wide TCEQ NO_2 mean over the period was 7.4483 ppb and co-located ozone at the same time was only 0.016 ppm, a pattern consistent with freshly emitted NO titration rather than regionally aged air.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The tceq NO_2 anomaly had an observed value of 9.1 ppb compared with an expected value of 1.6324324324324324 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

A shallow planetary boundary layer during the early morning hours restricted vertical dispersion, causing surface accumulation of nitrogen dioxide measured by TCEQ.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The NO2 anomaly recorded at 10:00 UTC on June 3, 2026, produced a z-score of 6.62 relative to baseline expectations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Morning meteorology near the anomaly featured very high humidity, extensive cloud cover, and a relatively shallow planetary boundary layer, which would favor reduced vertical dilution of near-surface NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Short-range transport from industrial or ship-channel emission areas is a plausible cause of the NO2 anomaly because surface winds near the event were from northeasterly to easterly directions while the affected monitor is in eastern Houston, allowing advection of an NOx plume from nearby source corridors toward the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The TCEQ NO2 monitor at the anomaly location recorded 9.1 ppb at 2026-06-03T10:00:00+00:00 versus an expected 1.63 ppb, which is most consistent with a monitor-scale fresh NOx plume rather than a regionally uniform enhancement.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

A severe NO2 anomaly of 9.1 ppb was detected by a TCEQ surface monitoring station in Houston at 10:00 UTC on June 3, 2026, exceeding the expected baseline of 1.63 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The PM2.5 levels were elevated due to the presence of pollutants in the atmosphere, likely originating from industrial or vehicular sources.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

Transport from nearby industrial or ship-channel combustion sources is plausible but not strongly confirmed because low-level winds near the anomaly were northeasterly at about 20 to 32 degrees around the event, which could advect emissions toward the eastern Houston site, but sulfur and broad regional combustion indicators were weak with TCEQ SO2 only 0.5 ppb at 10:00 UTC and sentinel5p SO2 near zero on the nearest overpass.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

The anomaly occurred on June 3rd, around 10:00 AM CDT, and was located approximately 8.161 km from the query point.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

Suppressed dispersion is a plausible cause of the NO_2 anomaly because humidity near the event was very high, morning cloud cover was extensive, and modeled planetary boundary layer height remained low, conditions that favor accumulation of primary NO_2 near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

High morning cloud cover and early-day light conditions limited solar photolysis, slowing the chemical loss of surface nitrogen dioxide.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

A severe thunderstorm or tornado event has stirred up pollutants and particles from the ground, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.